1) a)

$z^{-1} = (\cos\theta + \mathrm{i}\sin\theta)^{-1} = \cos(-\theta) + \mathrm{i}\sin(-\theta) = \cos\theta - \mathrm{i}\sin\theta$

Alternatively:

$z^{-1} = \frac{1}{\cos\theta + \mathrm{i}\sin\theta} = \frac{1}{\cos\theta + \mathrm{i}\sin\theta} \times \frac{\cos\theta - \mathrm{i}\sin\theta}{\cos\theta - \mathrm{i}\sin\theta} = \frac{\cos\theta - \mathrm{i}\sin\theta}{\cos^2\theta + \sin^2\theta} = \frac{\cos\theta - \mathrm{i}\sin\theta}{1} = \cos\theta - \mathrm{i}\sin\theta$

1) b)

$z + z^{-1} = \cos\theta + \mathrm{i}\sin\theta + \cos\theta - \mathrm{i}\sin\theta = 2\cos\theta$

So that $2\cos\theta = z + z^{-1}$ as req.

$z - z^{-1} = \cos\theta + \mathrm{i}\sin\theta - \cos\theta + \mathrm{i}\sin\theta = 2\mathrm{i}\sin\theta$

So that $2\sin\theta = z - z^{-1}$ as req.

2) a)

$z^{-n} = (\cos\theta + \mathrm{i}\sin\theta)^{-n} = \cos(-n\theta) + \mathrm{i}\sin(-n\theta) = \cos n\theta - \mathrm{i}\sin n\theta$

2) b)

$z + z^{-n} = \cos n\theta + \mathrm{i}\sin n\theta + \cos n\theta - \mathrm{i}\sin n\theta = 2\cos n\theta$

So that $\cos n\theta = \frac{z^n + z^{-n}}{2}$ as req.

2) c)

$z - z^{-1} = \cos n\theta + \mathrm{i}\sin n\theta - \cos n\theta + \mathrm{i}\sin n\theta = 2\mathrm{i}\sin n\theta$

So that $\sin n\theta = \frac{z^n - z^{-n}}{2}$ as req.

3) a)

$\left(z + \frac{1}{z}\right)^2 = z^2 + 2z \times \frac{1}{z} + \frac{1}{z^2} = z^2 + 2 + \frac{1}{z^2}$

3) b)

$\left(z + \frac{1}{z}\right)^3 = z^3 + 3z^2 \times \frac{1}{z} + 3z \times \frac{1}{z^2} + \frac{1}{z^3} = z^3 + 3z + \frac{3}{z} + \frac{1}{z^3}$

3) c)

$\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^3 \times \frac{1}{z} + 6z^2 \times \frac{1}{z^2} + 4z \times \frac{1}{z^3} + \frac{1}{z^4}$

$= z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}$

3) d)

$$\left(z+\frac{1}{z}\right)^5 = z^5 + 5z^4 \times \frac{1}{z} + 10z^3 \times \frac{1}{z^2} + 10z^2 \times \frac{1}{z^3} + 5z \times \frac{1}{z^4} + \frac{1}{z^5}$$
$$= z^5 + 5z^3 + 10z + \frac{10}{z} + \frac{5}{z^3} + \frac{1}{z^5}$$

3) e)

$$\left(z+\frac{1}{z}\right)^6 = z^6 + 6z^5 \times \frac{1}{z} + 15z^4 \times \frac{1}{z^2} + 20z^3 \times \frac{1}{z^3} + 15z^2 \times \frac{1}{z^4} + 6z \times \frac{1}{z^5} + \frac{1}{z^6}$$
$$= z^6 + 6z^4 + 15z^2 + 20 + \frac{15}{z^2} + \frac{6}{z^4} + \frac{1}{z^6}$$

4) a)

$$\left(z-\frac{1}{z}\right)^2 = z^2 - 2z \times \frac{1}{z} + \frac{1}{z^2} = z^2 - 2 + \frac{1}{z^2}$$

4) b)

$$\left(z-\frac{1}{z}\right)^3 = z^3 - 3z^2 \times \frac{1}{z} + 3z \times \frac{1}{z^2} - \frac{1}{z^3} = z^3 - 3z + \frac{3}{z} - \frac{1}{z^3}$$

4) c)

$$\left(z-\frac{1}{z}\right)^4 = z^4 - 4z^3 \times \frac{1}{z} + 6z^2 \times \frac{1}{z^2} - 4z \times \frac{1}{z^3} + \frac{1}{z^4}$$
$$= z^4 - 4z^2 + 6 - \frac{4}{z^2} + \frac{1}{z^4}$$

4) d)

$$\left(z-\frac{1}{z}\right)^5 = z^5 - 5z^4 \times \frac{1}{z} + 10z^3 \times \frac{1}{z^2} - 10z^2 \times \frac{1}{z^3} + 5z \times \frac{1}{z^4} - \frac{1}{z^5}$$
$$= z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5}$$

4) e)

$$\left(z-\frac{1}{z}\right)^6 = z^6 - 6z^5 \times \frac{1}{z} + 15z^4 \times \frac{1}{z^2} - 20z^3 \times \frac{1}{z^3} + 15z^2 \times \frac{1}{z^4} - 6z \times \frac{1}{z^5} + \frac{1}{z^6}$$
$$= z^6 - 6z^4 + 15z^2 - 20 + \frac{15}{z^2} - \frac{6}{z^4} + \frac{1}{z^6}$$

5) a)

Since $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$

$e^{i\theta} + e^{-i\theta} = 2\cos\theta$

So $\cos\theta = \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right)$

5) b)

Since $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$

$e^{i\theta} - e^{-i\theta} = 2i\sin\theta$

So $\sin\theta = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)$

6) a)

$\left(e^{i\theta} + e^{-i\theta}\right)^2 = e^{2i\theta} + 2e^{i\theta}e^{-i\theta} + e^{-2i\theta} = e^{2i\theta} + e^{-2i\theta} + 2$

6) b)

$\left(e^{i\theta} + e^{-i\theta}\right)^3 = e^{3i\theta} + 3e^{2i\theta}e^{-i\theta} + 3e^{i\theta}e^{-2i\theta} + e^{-3i\theta} = e^{3i\theta} + e^{-3i\theta} + 3e^{i\theta} + 3e^{-i\theta}$

6) c)

$\left(e^{i\theta} + e^{-i\theta}\right)^4 = e^{4i\theta} + 4e^{3i\theta}e^{-i\theta} + 6e^{2i\theta}e^{-2i\theta} + 4e^{i\theta}e^{-3i\theta} + e^{-4i\theta}$

$= e^{4i\theta} + e^{-4i\theta} + 4e^{2i\theta} + 4e^{-2i\theta} + 6$

7) a)

$\left[\frac{1}{i}\left(e^{i\theta} - e^{-i\theta}\right)\right]^2 = \left(\frac{1}{i}\right)^2\left(e^{i\theta} - e^{-i\theta}\right)^2 = -\left(e^{2i\theta} - 2e^{i\theta}e^{-i\theta} + e^{-2i\theta}\right) = 2 - e^{2i\theta} - e^{-2i\theta}$

7) b)

$\left[\frac{1}{i}\left(e^{i\theta} - e^{-i\theta}\right)\right]^3 = \left(\frac{1}{i}\right)^3\left(e^{i\theta} - e^{-i\theta}\right)^3 = i\left(e^{3i\theta} - 3e^{2i\theta}e^{-i\theta} + 3e^{i\theta}e^{-2i\theta} - e^{-3i\theta}\right)$

$= i\left(e^{3i\theta} - 3e^{i\theta} + 3e^{-i\theta} - e^{-3i\theta}\right)$

7) c)

$\left[\frac{1}{i}\left(e^{i\theta} - e^{-i\theta}\right)\right]^4 = \left(\frac{1}{i}\right)^4\left(e^{i\theta} - e^{-i\theta}\right)^4$

$= e^{4i\theta} - 4e^{3i\theta}e^{-i\theta} + 6e^{2i\theta}e^{-2i\theta} - 4e^{i\theta}e^{-3i\theta} + e^{-4i\theta}$

$= e^{4i\theta} + e^{-4i\theta} - 4e^{2i\theta} - 4e^{-2i\theta} + 6$

8)

$\left(z+\frac{1}{z}\right)^2 = z^2+2+\frac{1}{z^2} = \left(z^2+\frac{1}{z^2}\right)+2$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$\left(z+\frac{1}{z}\right)^2 = 2\cos 2\theta + 2$

So $(2\cos\theta)^2 = 2\cos 2\theta + 2$

$\therefore \cos^2\theta = \frac{1}{2}\cos 2\theta + \frac{1}{2}$

9)

$\left(z-\frac{1}{z}\right)^2 = z^2-2+\frac{1}{z^2} = \left(z^2+\frac{1}{z^2}\right)-2$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$\left(z-\frac{1}{z}\right)^2 = 2\cos 2\theta - 2$

So $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ gives

$(2\mathrm{i}\sin\theta)^2 = 2\cos 2\theta - 2$

$-4\sin\theta = 2\cos 2\theta - 2$

$\therefore \sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$ as req.

10) a)

Since $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ and $\mathrm{e}^{-\mathrm{i}\theta} = \cos\theta - \mathrm{i}\sin\theta$

$\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta} = 2\cos\theta$

So $\cos\theta = \frac{1}{2}\left(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}\right)$

$\therefore \cos^3\theta = \left[\frac{1}{2}\left(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}\right)\right]^3 = \frac{1}{8}\left(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}\right)^3$

$= \frac{1}{8}\left(\mathrm{e}^{3\mathrm{i}\theta} + 3\mathrm{e}^{-\mathrm{i}\theta} + 3\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-3\mathrm{i}\theta}\right)$

$= \frac{1}{8}\left(\mathrm{e}^{3\mathrm{i}\theta} + \mathrm{e}^{-3\mathrm{i}\theta} + 3\left(\mathrm{e}^{-\mathrm{i}\theta} + \mathrm{e}^{\mathrm{i}\theta}\right)\right)$

$= \frac{1}{8}\left(2\cos 3\theta + 3(2\cos\theta)\right)$

$= \frac{1}{4}\cos 3\theta + \frac{3}{4}\cos\theta$

10) b)

$$\int_0^{\frac{\pi}{3}} \cos^3\theta \,\mathrm{d}x = \int_0^{\frac{\pi}{3}} \frac{1}{4}\cos(3\theta) + \frac{3}{4}\cos\theta \,\mathrm{d}x = \left[\frac{1}{12}\sin(3\theta) + \frac{3}{4}\sin\theta\right]_0^{\frac{\pi}{3}}$$

$$= \left(0 + \frac{3\sqrt{3}}{8}\right) - (0+0) = \frac{3\sqrt{3}}{8}$$

So $p = 3$ and $q = 8$

11) a)

$$\left(z - \frac{1}{z}\right)^3 = z^3 - 3z + \frac{3}{z} - \frac{1}{z^3} = \left(z^3 - \frac{1}{z^3}\right) - 3\left(z - \frac{1}{z}\right)$$

Since $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ then

$$\left(z + \frac{1}{z}\right)^3 = 2\mathrm{i}\sin 3\theta - 3 \times 2\mathrm{i}\sin\theta$$

So $(2\mathrm{i}\sin\theta)^3 = 2\mathrm{i}\sin 3\theta - 6\mathrm{i}\sin\theta$

$-8\mathrm{i}\sin^3\theta = 2\mathrm{i}\sin 3\theta - 6\mathrm{i}\sin\theta$

$\therefore \sin^3\theta = -\frac{1}{4}\sin 3\theta + \frac{3}{4}\sin\theta$ as req.

11) b)

$\sin^3\theta = \frac{1}{8} \Rightarrow \sin\theta = \frac{1}{2}$

So $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

12) a)

$$\left(z - \frac{1}{z}\right)^4 = z^4 - 4z^2 + 6 - \frac{4}{z^2} + \frac{1}{z^4} = \left(z^4 + \frac{1}{z^4}\right) - 4\left(z^2 + \frac{1}{z^2}\right) + 6$$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$$\left(z - \frac{1}{z}\right)^4 = 2\cos 4\theta - 4 \times 2\cos 2\theta + 6$$

Since $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ then

So $(2\mathrm{i}\sin\theta)^4 = 2\cos 4\theta - 8\cos 2\theta + 6$

$16\sin^4\theta = 2\cos 4\theta - 8\cos 2\theta + 6$

$\therefore \sin^4\theta = \frac{1}{8}\cos 4\theta - \frac{1}{2}\cos 2\theta + \frac{3}{8} = \frac{1}{8}(\cos 4\theta - 4\cos 2\theta + 3)$

So $a = 1$, $b = -4$ and $c = 3$

12) b)

Let $\theta = \frac{\pi}{8}$ so

$\sin^4\left(\frac{\pi}{8}\right) = \frac{1}{8}\left(\cos\left(4 \times \frac{\pi}{8}\right) - 4\cos\left(2 \times \frac{\pi}{8}\right) + 3\right)$

$= \frac{1}{8}\left(\cos\left(\frac{\pi}{2}\right) - 4\cos\left(\frac{\pi}{4}\right) + 3\right)$

$= \frac{1}{8}\left(0 - 4 \times \frac{\sqrt{2}}{2} + 3\right) = \frac{1}{8}\left(3 - 2\sqrt{2}\right)$

$\therefore \sin\left(\frac{\pi}{8}\right) = \left(\frac{3-2\sqrt{2}}{8}\right)^{\frac{1}{4}}$ as req.

13) a)

$\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4} = \left(z^4 + \frac{1}{z^4}\right) + 4\left(z^2 + \frac{1}{z^2}\right) + 6$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$\left(z + \frac{1}{z}\right)^4 = 2\cos 4\theta + 4 \times 2\cos 2\theta + 6$

So $(2\cos\theta)^4 = 2\cos 4\theta + 8\cos 2\theta + 6$

$\therefore \cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}$

13) b)

$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} 8\cos^4\theta \ \mathrm{d}\theta = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos 4\theta + 4\cos 2\theta + 3 \ \mathrm{d}\theta$

$= \left[\frac{1}{4}\sin 4\theta + 2\sin 2\theta + 3\theta\right]_{\frac{\pi}{2}}^{\frac{3\pi}{2}} = \left(0 + 0 + \frac{9\pi}{2}\right) - \left(0 + 0 + \frac{3\pi}{2}\right) = 3\pi$

14) a)

He incorrectly defines $2\sin\theta$ as $\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}$ but $\sin\theta = \frac{1}{2\mathrm{i}}\left(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta}\right)$

14) b)

$16\sin^4\theta \equiv (2\sin\theta)^4$

$\equiv \left(2 \times \frac{1}{2\mathrm{i}}\left(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta}\right)\right)^4$

$\equiv \left(-\mathrm{i}\left(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta}\right)\right)^4$

$\equiv (-\mathrm{i})^4\left(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta}\right)^4$

$\equiv \mathrm{e}^{4\mathrm{i}\theta} - 4\mathrm{e}^{2\mathrm{i}\theta} + 6 - 4\mathrm{e}^{-2\mathrm{i}\theta} + \mathrm{e}^{-4\mathrm{i}\theta}$

$\equiv \mathrm{e}^{4\mathrm{i}\theta} + \mathrm{e}^{-4\mathrm{i}\theta} - 4\left(\mathrm{e}^{2\mathrm{i}\theta} + \mathrm{e}^{-2\mathrm{i}\theta}\right) + 6$

$\equiv 2\cos 4\theta - 8\cos 2\theta + 6$

So that $32\sin^4\theta \equiv 4\cos 4\theta - 16\cos 2\theta + 12$ as req.

14) c)

$16\cos^4\theta \equiv (2\cos\theta)^4$

$\equiv \left(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}\right)^4$

$\equiv \mathrm{e}^{4\mathrm{i}\theta} + 4\mathrm{e}^{2\mathrm{i}\theta} + 6 + 4\mathrm{e}^{-2\mathrm{i}\theta} + \mathrm{e}^{-4\mathrm{i}\theta}$

$\equiv \mathrm{e}^{4\mathrm{i}\theta} + \mathrm{e}^{-4\mathrm{i}\theta} + 4\left(\mathrm{e}^{2\mathrm{i}\theta} + \mathrm{e}^{-2\mathrm{i}\theta}\right) + 6$

$\equiv 2\cos 4\theta + 8\cos 2\theta + 6$

So that $32\cos^4\theta \equiv 4\cos 4\theta + 16\cos 2\theta + 12$ as req.

15) a)

Since $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ and $\mathrm{e}^{-\mathrm{i}\theta} = \cos\theta - \mathrm{i}\sin\theta$

$\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta} = 2\cos\theta$

So $\cos\theta = \frac{1}{2}\left(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}\right)$

$\therefore \cos^5\theta = \left[\frac{1}{2}\left(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}\right)\right]^5 = \frac{1}{32}\left(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}\right)^5$

$= \frac{1}{32}\left(\mathrm{e}^{5\mathrm{i}\theta} + 5\mathrm{e}^{4\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\theta} + 10\mathrm{e}^{3\mathrm{i}\theta}\mathrm{e}^{-2\mathrm{i}\theta} + 10\mathrm{e}^{2\mathrm{i}\theta}\mathrm{e}^{-3\mathrm{i}\theta} + 5\mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-4\mathrm{i}\theta} + \mathrm{e}^{-5\mathrm{i}\theta}\right)$

$= \frac{1}{32}\left(\mathrm{e}^{5\mathrm{i}\theta} + \mathrm{e}^{-5\mathrm{i}\theta} + 5\left(\mathrm{e}^{3\mathrm{i}\theta} + \mathrm{e}^{-3\mathrm{i}\theta}\right) + 10\left(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}\right)\right)$

$= \frac{1}{32}\left(2\cos 5\theta + 5(2\cos 3\theta) + 10(2\cos\theta)\right)$

$= \frac{1}{16}\cos 5\theta + \frac{5}{16}\cos 3\theta + \frac{5}{8}\cos\theta$

15) b)

$2\cos 5\theta + 10\cos 3\theta + 20\cos\theta = 1$

$\Rightarrow \frac{1}{16}\cos 5\theta + \frac{5}{16}\cos 3\theta + \frac{5}{8}\cos\theta = \frac{1}{32}$

$\Rightarrow \cos^5\theta = \frac{1}{32}$

$\Rightarrow \cos\theta = \left(\frac{1}{32}\right)^{\frac{1}{5}}$

$\therefore \theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$

16) a)

$\left(z - \frac{1}{z}\right)^5 = z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5} = \left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right)$

Since $z^n + z^{-n} = 2\mathrm{i}\sin n\theta$ then

$\left(z - \frac{1}{z}\right)^5 = 2\mathrm{i}\sin 5\theta - 5 \times 2\mathrm{i}\sin 3\theta + 10 \times 2\mathrm{i}\sin\theta$

So $(2\mathrm{i}\sin\theta)^5 = 2\mathrm{i}\sin 5\theta - 10\mathrm{i}\sin 3\theta + 20\mathrm{i}\sin\theta$

$32\mathrm{i}\sin^5\theta = 2\mathrm{i}\sin 5\theta - 10\mathrm{i}\sin 3\theta + 20\mathrm{i}\sin\theta$

$\therefore \sin^5\theta = \frac{1}{16}\sin 5\theta - \frac{5}{16}\sin 3\theta + \frac{5}{8}\sin\theta$

16) b)

$\int_0^{\frac{\pi}{2}} 16\sin^5\theta \ \mathrm{d}\theta = \int_0^{\frac{\pi}{2}} \sin 5\theta - 5\sin 3\theta + 10\sin\theta \ \mathrm{d}\theta$

$= \left[-\frac{1}{5}\cos 5\theta + \frac{5}{3}\cos 3\theta - 10\cos\theta\right]_0^{\frac{\pi}{2}} = (0 + 0 + 0) - \left(-\frac{1}{5} + \frac{5}{3} - 10\right) = \frac{128}{15}$

17) a)

$\left(z + \frac{1}{z}\right)^6 = z^6 + 6z^4 + 15z^2 + 20 + \frac{15}{z^2} + \frac{6}{z^4} + \frac{1}{z^6}$

$= \left(z^6 + \frac{1}{z^6}\right) + 6\left(z^4 + \frac{1}{z^4}\right) + 15\left(z^2 + \frac{1}{z^2}\right) + 20$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$\left(z + \frac{1}{z}\right)^6 = 2\cos 6\theta + 6 \times 2\cos 4\theta + 15 \times 2\cos 2\theta + 20$

So $(2\cos\theta)^6 = 2\cos 6\theta + 12\cos 4\theta + 30\cos 2\theta + 20$

$\therefore \cos^6\theta = \frac{1}{32}\cos 6\theta + \frac{3}{16}\cos 4\theta + \frac{15}{32}\cos 2\theta + \frac{5}{16}$

17) b)

Let $\theta = \frac{\pi}{12}$ so

$\cos^6\left(\frac{\pi}{12}\right) = \frac{1}{32}\cos\left(6 \times \frac{\pi}{12}\right) + \frac{3}{16}\cos\left(4 \times \frac{\pi}{12}\right) + \frac{15}{32}\cos\left(2 \times \frac{\pi}{12}\right) + \frac{5}{16}$

$= \frac{1}{32}\cos\frac{\pi}{2} + \frac{3}{16}\cos\frac{\pi}{3} + \frac{15}{32}\cos\frac{\pi}{6} + \frac{5}{16}$

$= 0 + \frac{3}{32} + \frac{15\sqrt{3}}{64} + \frac{5}{16} = \frac{13}{32} + \frac{15\sqrt{3}}{64} = \frac{26+15\sqrt{3}}{64}$

$\therefore \cos\left(\frac{\pi}{12}\right) = \left(\frac{26+15\sqrt{3}}{64}\right)^{\frac{1}{6}} = \frac{1}{2}\left(26 + 15\sqrt{3}\right)^{\frac{1}{6}}$ as req.

18) a)

Since $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$

$e^{i\theta} - e^{-i\theta} = 2i\sin\theta$

So $\sin\theta = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)$

$\therefore \sin^6\theta = \left[\frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)\right]^6 = -\frac{1}{64}\left(e^{i\theta} - e^{-i\theta}\right)^6$

$= -\frac{1}{64}\left(e^{6i\theta} - 6e^{5i\theta}e^{-i\theta} + 15e^{4i\theta}e^{-2i\theta} - 20e^{3i\theta}e^{-3i\theta} + 15e^{2i\theta}e^{-4i\theta} - 6e^{i\theta}e^{-5i\theta} + e^{-6i\theta}\right)$

$= -\frac{1}{64}\left(e^{6i\theta} + e^{-6i\theta} - 6\left(e^{4i\theta} + e^{-4i\theta}\right) + 15\left(e^{2i\theta} + e^{-2i\theta}\right) - 20\right)$

$= -\frac{1}{64}\left(2\cos 6\theta - 6(2\cos 4\theta) + 15(2\cos 2\theta) - 20\right)$

$= -\frac{1}{32}\cos 6\theta + \frac{3}{16}\cos 4\theta - \frac{15}{32}\cos 2\theta + \frac{5}{16}$

$\therefore 32\sin^6\theta = -\cos 6\theta + 6\cos 4\theta - 15\cos 2\theta + 10$

18) b)

Let $\theta = \frac{\pi}{12}$ so

$\sin^6\left(\frac{\pi}{12}\right) = -\frac{1}{32}\cos\left(6 \times \frac{\pi}{12}\right) + \frac{3}{16}\cos\left(4 \times \frac{\pi}{12}\right) - \frac{15}{32}\cos\left(2 \times \frac{\pi}{12}\right) + \frac{5}{16}$

$= -\frac{1}{32}\cos\left(\frac{\pi}{2}\right) + \frac{3}{16}\cos\left(\frac{\pi}{3}\right) - \frac{15}{32}\cos\left(\frac{\pi}{6}\right) + \frac{5}{16}$

$= -\frac{1}{32}\cos\frac{\pi}{2} + \frac{3}{16}\cos\frac{\pi}{3} - \frac{15}{32}\cos\frac{\pi}{6} + \frac{5}{16}$

$= 0 + \frac{3}{32} - \frac{15\sqrt{3}}{64} + \frac{5}{16} = \frac{13}{32} - \frac{15\sqrt{3}}{64} = \frac{26-15\sqrt{3}}{64}$

$\therefore \sin\left(\frac{\pi}{12}\right) = \left(\frac{26-15\sqrt{3}}{64}\right)^{\frac{1}{6}} = \frac{1}{2}\left(26 - 15\sqrt{3}\right)^{\frac{1}{6}}$ as req.

19)

$\left(z^3+\frac{1}{z^3}\right)^2 = z^6+2+\frac{1}{z^6} = \left(z^6+\frac{1}{z^6}\right)+2$

Since $z^n+z^{-n} = 2\cos n\theta$ then

$\left(z^3+\frac{1}{z^3}\right)^2 = 2\cos 6\theta + 2$

But also

$\left(z^3+\frac{1}{z^3}\right)^2 = (2\cos 3\theta)^2 = 4\cos^2 3\theta$

So $4\cos^2 3\theta = 2\cos 6\theta + 2$

$\therefore 2\cos^2 3\theta = \cos 6\theta + 1$

20)

$\left(z^4-\frac{1}{z^4}\right)^3 = z^{12}-3z^4+\frac{3}{z^4}-\frac{1}{z^{12}} = \left(z^{12}-\frac{1}{z^{12}}\right)-3\left(z^4-\frac{1}{z^4}\right)$

Since $z^n-z^{-n} = 2\mathrm{i}\sin n\theta$ then

$\left(z^4-\frac{1}{z^4}\right)^3 = 2\mathrm{i}\sin 12\theta - 6\mathrm{i}\sin 4\theta$

But also

$\left(z^4-\frac{1}{z^4}\right)^3 = (2\mathrm{i}\sin 4\theta)^3 = -8\mathrm{i}\sin^3 4\theta$

So $-8\mathrm{i}\sin^3 4\theta = 2\mathrm{i}\sin 12\theta - 6\mathrm{i}\sin 4\theta$

$\therefore 4\sin^3 4\theta = -\sin 12\theta + 3\sin 4\theta$ as req.

21)

$\left(z^3-\frac{1}{z^3}\right)^5 = z^{15}-5z^9+10z^3-\frac{10}{z^3}+\frac{5}{z^9}-\frac{1}{z^{15}}$

$= \left(z^{15}-\frac{1}{z^{15}}\right)-5\left(z^9-\frac{1}{z^9}\right)+10\left(z^3-\frac{1}{z^3}\right)$

Since $z^n-z^{-n} = 2\mathrm{i}\sin n\theta$ then

$\left(z^3-\frac{1}{z^3}\right)^5 = 2\mathrm{i}\sin 15\theta - 10\mathrm{i}\sin 9\theta + 20\mathrm{i}\sin 3\theta$

But also

$\left(z^3-\frac{1}{z^3}\right)^5 = (2\mathrm{i}\sin 3\theta)^5 = 32\mathrm{i}\sin^5 3\theta$

So $32\mathrm{i}\sin^5 3\theta = 2\mathrm{i}\sin 15\theta - 10\mathrm{i}\sin 9\theta + 20\mathrm{i}\sin 3\theta$

$\therefore \sin^5 3\theta = \frac{1}{16}\sin 15\theta - \frac{5}{16}\sin 9\theta + \frac{5}{8}\sin 3\theta$

22) a)

$z^n = \cos n\theta + \mathrm{i}\sin n\theta$

$z^{-n} = \cos(-n\theta) + \mathrm{i}\sin(-n\theta) = \cos n\theta - \mathrm{i}\sin n\theta$

$z^n + z^{-n} = \cos n\theta + \mathrm{i}\sin n\theta + \cos n\theta - \mathrm{i}\sin n\theta = 2\cos n\theta$

22) b)

$z^n - z^{-n} = \cos n\theta + \mathrm{i}\sin n\theta - \cos n\theta + \mathrm{i}\sin n\theta = 2\mathrm{i}\sin n\theta$

22) c)

$\left(z+\frac{1}{z}\right)^2\left(1-\frac{1}{z}\right)^2 = \left(z^2+1+1+\frac{1}{z^2}\right)\left(z^2-1-1+\frac{1}{z^2}\right)$

$= \left(z^2+2+\frac{1}{z^2}\right)\left(z^2-2+\frac{1}{z^2}\right) = z^4 - 2z^2 + 1 + 2z^2 - 4 + \frac{2}{z^2} + 1 - \frac{2}{z^2} + \frac{1}{z^4}$

$= \left(z^4+\frac{1}{z^4}\right) - 2$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$\left(z+\frac{1}{z}\right)^2\left(1-\frac{1}{z}\right)^2 = 2\cos 4\theta - 2$

But also

$\left(z+\frac{1}{z}\right)^2\left(1-\frac{1}{z}\right)^2 = (2\cos\theta)^2(2\mathrm{i}\sin\theta)^2 = -16\cos^2\theta\sin^2\theta$

So $-16\cos^2\theta\sin^2\theta = 2\cos 4\theta - 2$

$\therefore \cos^2\theta\sin^2\theta = -\frac{2}{16}\cos 4\theta + \frac{2}{16} = \frac{1}{8}(1-\cos 4\theta)$ as req.

23) a)

$z^n = \cos n\theta + \mathrm{i}\sin n\theta$

$z^{-n} = \cos(-n\theta) + \mathrm{i}\sin(-n\theta) = \cos n\theta - \mathrm{i}\sin n\theta$

$z^n + z^{-n} = \cos n\theta + \mathrm{i}\sin n\theta + \cos n\theta - \mathrm{i}\sin n\theta = 2\cos n\theta$

23) b)

$z^n - z^{-n} = \cos n\theta + \mathrm{i}\sin n\theta - \cos n\theta + \mathrm{i}\sin n\theta = 2\mathrm{i}\sin n\theta$

23) c)

$\left(z+\frac{1}{z}\right)^3\left(z-\frac{1}{z}\right)^3=\left(z^3+3z+\frac{3}{z}+\frac{1}{z^3}\right)\left(z^3-3z+\frac{3}{z}-\frac{1}{z^3}\right)$

$=z^6-3z^4+3z^2-1+3z^4-9z^2+9-\frac{3}{z^2}+3z^2-9+\frac{9}{z^2}-\frac{3}{z^4}+1-\frac{3}{z^2}+\frac{3}{z^4}-\frac{1}{z^6}$

$=\left(z^6-\frac{1}{z^6}\right)-3\left(z^2-\frac{3}{z^2}\right)$

Since $z^n-z^{-n}=2\mathrm{i}\sin n\theta$ then

$\left(z+\frac{1}{z}\right)^3\left(1-\frac{1}{z}\right)^3=2\mathrm{i}\sin 6\theta-6\mathrm{i}\sin 2\theta$

But also

$\left(z+\frac{1}{z}\right)^3\left(1-\frac{1}{z}\right)^3=(2\cos\theta)^3(2\mathrm{i}\sin\theta)^3=-64\mathrm{i}\cos^3\theta\sin^3\theta$

So $-64\mathrm{i}\cos^3\theta\sin^3\theta=2\mathrm{i}\sin 6\theta-6\mathrm{i}\sin 2\theta$

$\therefore\cos^3\theta\sin^3\theta=-\frac{1}{32}\sin 6\theta+\frac{3}{32}\sin 2\theta=\frac{1}{32}(3\sin 2\theta-\sin 6\theta)$ as req.

24)

Let $z=\cos\theta+\mathrm{i}\sin\theta$

$\left(z+\frac{1}{z}\right)^3\left(z-\frac{1}{z}\right)^4=\left(z^3+3z+\frac{3}{z}+\frac{1}{z^3}\right)\left(z^4-4z^2+6-\frac{4}{z^2}+\frac{1}{z^4}\right)$

$=z^7+3z^5+3z^3+z+-4z^5-12z^3-12z-\frac{4}{z}+6z^3+18z+\frac{18}{z}+\frac{6}{z^3}-\frac{4}{z^5}-\frac{12}{z^3}-4z-\frac{12}{z}+\frac{1}{z}+\frac{3}{z^3}+\frac{3}{z^5}+\frac{1}{z^7}$

$=\left(z^7+\frac{1}{z^7}\right)+\left(-z^5-\frac{1}{z^5}\right)+\left(-3z^3-\frac{3}{z^3}\right)+\left(3z+\frac{3}{z}\right)$

$=\left(z^7+\frac{1}{z^7}\right)-\left(z^5+\frac{1}{z^5}\right)-\left(3z^3+\frac{3}{z^3}\right)+\left(3z+\frac{3}{z}\right)$

So that since $z^n+z^{-n}=2\cos n\theta$ and $z^n-z^{-n}=2\mathrm{i}\sin n\theta$ then

$(2\cos\theta)^3(2\mathrm{i}\sin\theta)^4=2\cos 7\theta-2\cos 5\theta-3\times 2\cos 3\theta+3\times 2\cos\theta$

$8\cos^3\theta\times 16\sin^4\theta=2\cos 7\theta-2\cos 5\theta-6\cos 3\theta+6\cos\theta$

$\cos^3\theta\sin^4\theta=\frac{1}{64}\cos 7\theta-\frac{1}{64}\cos 5\theta-\frac{3}{64}\cos 3\theta+\frac{3}{64}\cos\theta$

$=\frac{1}{64}(\cos 7\theta-\cos 5\theta-3\cos 3\theta+3\cos\theta)$

25) a)

$\mathrm{Re}\left(\mathrm{e}^{3\mathrm{i}\theta}\left(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta}\right)^3\right)=\mathrm{Re}\left((\cos 3\theta+\mathrm{i}\sin 3\theta)(2\cos\theta)^3\right)$

$=8\cos 3\theta\cos^3\theta$ so $k=8$

25) b)

$\left(e^{i\theta}+e^{-i\theta}\right)^3 = e^{3i\theta}+3e^{i\theta}+3e^{-i\theta}+e^{-3i\theta}$

$\therefore e^{3i\theta}\left(e^{i\theta}+e^{-i\theta}\right)^3 = e^{3i\theta}\left(e^{3i\theta}+3e^{i\theta}+3e^{-i\theta}+e^{-3i\theta}\right)$

$= e^{6i\theta}+3e^{4i\theta}+3e^{2i\theta}+1$

So $\text{Re}\left(e^{6i\theta}+3e^{4i\theta}+3e^{2i\theta}+1\right) = \cos 6\theta+3\cos 4\theta+3\cos 2\theta+1$

$\therefore 8\cos 3\theta\cos^3\theta = \cos 6\theta+3\cos 4\theta+3\cos 2\theta+1$

Let $\theta = \frac{\pi}{12}$

$8\cos\frac{\pi}{4}\cos^3\frac{\pi}{12} = \cos\frac{\pi}{2}+3\cos\frac{\pi}{3}+3\cos\frac{\pi}{6}+1$

$\Rightarrow 4\sqrt{2}\cos^3\frac{\pi}{12} = 0+\frac{3}{2}+\frac{3\sqrt{3}}{2}+1$

$\Rightarrow \cos^3\frac{\pi}{12} = \frac{5}{8\sqrt{2}}+\frac{3\sqrt{3}}{8\sqrt{2}} = \frac{5\sqrt{2}+3\sqrt{6}}{16} = \frac{10\sqrt{2}}{32}+\frac{3\sqrt{6}}{32}$

$\therefore \cos\frac{\pi}{12} = \sqrt[3]{\frac{5\sqrt{2}+3\sqrt{6}}{16}}$

So $p = 5$ and $q = 3$

26) a)

$\text{Im}\left(e^{4i\theta}\left(e^{i\theta}-e^{-i\theta}\right)^4\right) = \text{Im}\left((\cos 4\theta+i\sin 4\theta)(2i\sin\theta)^4\right)$

$= 16\sin 4\theta\sin^4\theta$ so $k = 16$

26) b)

$\left(e^{i\theta}-e^{-i\theta}\right)^4 = e^{4i\theta}-4e^{2i\theta}+6-4e^{-2i\theta}+e^{-4i\theta}$

$\therefore e^{4i\theta}\left(e^{i\theta}-e^{-i\theta}\right)^4 = e^{4i\theta}\left(e^{4i\theta}-4e^{2i\theta}+6-4e^{-2i\theta}+e^{-4i\theta}\right)$

$= e^{8i\theta}-4e^{6i\theta}+6e^{4i\theta}-4e^{2i\theta}+1$

So $\text{Im}\left(e^{8i\theta}-4e^{6i\theta}+6e^{4i\theta}-4e^{2i\theta}\right)$

$= \sin 8\theta-4\sin 6\theta+6\sin 4\theta-4\sin 2\theta$

$\therefore 16\sin 4\theta\sin^4\theta = \sin 8\theta-4\sin 6\theta+6\sin 4\theta-4\sin 2\theta$

Let $\theta = \frac{\pi}{8}$

$16\sin\frac{\pi}{2}\sin^4\frac{\pi}{8} = \sin\pi-4\sin\frac{3\pi}{4}+6\sin\frac{\pi}{2}-4\sin\frac{\pi}{4}$

$\Rightarrow 16\sin^4\frac{\pi}{8} = 0-2\sqrt{2}+6-2\sqrt{2}$

$\Rightarrow \sin^4\frac{\pi}{8} = \frac{6-4\sqrt{2}}{16}$

$\therefore \sin\frac{\pi}{8} = \sqrt[4]{\frac{6-4\sqrt{2}}{16}} = \frac{1}{2}\left(6-4\sqrt{2}\right)^{\frac{1}{4}}$

So $p = 6$ and $q = 4$

27) a)

$$\int \sin\theta(\cos\theta)^4 \, \mathrm{d}\theta = -\frac{1}{5}\cos^5\theta + C$$

27) b)

$$\left(z+\tfrac{1}{z}\right)^4\left(z-\tfrac{1}{z}\right) = \left(z^4+4z^2+6+\tfrac{4}{z^2}+\tfrac{1}{z^4}\right)\left(z-\tfrac{1}{z}\right)$$
$$= z^5+4z^3+6z+\tfrac{4}{z}+\tfrac{1}{z^3}-z^3-4z-\tfrac{6}{z}-\tfrac{4}{z^3}-\tfrac{1}{z^5}$$
$$= \left(z^5-\tfrac{1}{z^5}\right)+3\left(z^3-\tfrac{1}{z^3}\right)+2\left(z-\tfrac{1}{z}\right)$$

Since $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ then

$$\left(z+\tfrac{1}{z}\right)^4\left(z-\tfrac{1}{z}\right) = 2\mathrm{i}\sin 5\theta + 6\mathrm{i}\sin 3\theta + 4\mathrm{i}\sin\theta$$

But also

$$\left(z+\tfrac{1}{z}\right)^4\left(z-\tfrac{1}{z}\right) = (2\cos\theta)^4 2\mathrm{i}\sin\theta = 32\mathrm{i}\cos^4\theta\sin\theta$$

So $32\mathrm{i}\cos^4\theta\sin\theta = 2\mathrm{i}\sin 5\theta + 6\mathrm{i}\sin 3\theta + 4\mathrm{i}\sin\theta$

$$\therefore \cos^4\theta\sin\theta = \frac{1}{16}\sin 5\theta + \frac{3}{16}\sin 3\theta + \frac{1}{8}\sin\theta$$

$$\int \sin\theta(\cos\theta)^4 \, \mathrm{d}\theta = \int \left(\frac{1}{16}\sin 5\theta + \frac{3}{16}\sin 3\theta + \frac{1}{8}\sin\theta\right) \mathrm{d}\theta$$
$$= -\frac{1}{80}\cos 5\theta - \frac{1}{16}\cos 3\theta - \frac{1}{8}\cos\theta + C$$

28) a)

$$\left(z-\tfrac{1}{z}\right)^3\left(z+\tfrac{1}{z}\right)^2 = \left(z^3-3z+\tfrac{3}{z}-\tfrac{1}{z^3}\right)\left(z^2+2+\tfrac{1}{z^2}\right)$$
$$= z^5+2z^3+z-3z^3-6z-\tfrac{3}{z}+3z+\tfrac{6}{z}+\tfrac{3}{z^3}-\tfrac{1}{z}-\tfrac{2}{z^3}-\tfrac{1}{z^5}$$
$$= \left(z^5-\tfrac{1}{z^5}\right)-\left(z^3-\tfrac{1}{z^3}\right)-2\left(z-\tfrac{1}{z}\right)$$

Since $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ then

$$\left(z-\tfrac{1}{z}\right)^3\left(z+\tfrac{1}{z}\right)^2 = 2\mathrm{i}\sin 5\theta - 2\mathrm{i}\sin 3\theta - 4\mathrm{i}\sin\theta$$

But also

$$\left(z-\tfrac{1}{z}\right)^3\left(z+\tfrac{1}{z}\right)^2 = (2\mathrm{i}\sin\theta)^3(2\cos\theta)^2 = -32\mathrm{i}\sin^3\theta\cos^2\theta$$

So $-32\mathrm{i}\sin^3\theta\cos^2\theta = 2\mathrm{i}\sin 5\theta - 2\mathrm{i}\sin 3\theta - 4\mathrm{i}\sin\theta$

$$\therefore 16\sin^3\theta\cos^2\theta = 2\sin\theta + \sin 3\theta - \sin 5\theta$$

So $a = 2$, $b = 1$ and $c = -1$

28) b)

Choose $x = \sin\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \cos\theta$ and when $x = 0$, $\theta = 0$, when $x = 1$, $\theta = \frac{\pi}{2}$

$$\int_0^{\frac{\pi}{2}} 16\sin^3\theta\sqrt{1-\sin^2\theta} \times \cos\theta\,\mathrm{d}\theta = \int_0^{\frac{\pi}{2}} 16\sin^3\theta\cos^2\theta\;\mathrm{d}\theta$$

$$= \int_0^{\frac{\pi}{2}} (2\sin\theta + \sin 3\theta - \sin 5\theta)\,\mathrm{d}\theta$$

$$= \left[-2\cos\theta - \frac{1}{3}\cos 3\theta + \frac{1}{5}\cos 5\theta\right]_0^{\frac{\pi}{2}}$$

$= (0+0+0) - \left(-2 - \frac{1}{3} + \frac{1}{5}\right) = \frac{32}{15}$ as req.

29) a)

$$\left(z - \frac{1}{z}\right)^2\left(z + \frac{1}{z}\right)^2 = \left(z^2 - 2 + \frac{1}{z^2}\right)\left(z^2 + 2 + \frac{1}{z^2}\right)$$

$$= z^4 - 2z^2 + 1 + 2z^2 - 4 + \frac{2}{z^2} + 1 - \frac{2}{z^2} + \frac{1}{z^4}$$

$$= \left(z^4 + \frac{1}{z^4}\right) - 2$$

Since $z^n - z^{-n} = 2\cos n\theta$

$$\left(z - \frac{1}{z}\right)^2\left(z + \frac{1}{z}\right)^2 = 2\cos 4\theta - 2$$

But also

$$\left(z - \frac{1}{z}\right)^2\left(z + \frac{1}{z}\right)^2 = (2\mathrm{i}\sin\theta)^2(2\cos\theta)^2 = -16\sin^2\theta\cos^2\theta$$

So $-16\sin^2\theta\cos^2\theta = 2\cos 4\theta - 2$

$\therefore 8\sin^2\theta\cos^2\theta = 1 - \cos 4\theta$

So $a = 1$ and $b = -1$

29) b)

Choose $x = 2\sin\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = 2\cos\theta$ and when $x = 0$, $\theta = 0$, when $x = 2$, $\theta = \frac{\pi}{2}$

$$\int_0^{\frac{\pi}{2}} 4 \times 4\sin^2\theta\sqrt{4 - 4\sin^2\theta} \times 2\cos\theta\,\mathrm{d}\theta = \int_0^{\frac{\pi}{2}} 32\sin^2\theta \times 2\sqrt{1 - 1\sin^2\theta} \times \cos\theta\,\mathrm{d}\theta$$

$$= \int_0^{\frac{\pi}{2}} 64\sin^2\theta\cos^2\theta\,\mathrm{d}\theta$$

$$= \int_0^{\frac{\pi}{2}} (8 - 8\cos 4\theta)\mathrm{d}\theta$$

$$= [8\theta - 2\sin 4\theta]_0^{\frac{\pi}{2}}$$

$= (4\pi - 0) - (0 - 0) = 4\pi$ as req.

30) a)

$$\left(z - \frac{1}{z}\right)^6 = z^6 - 6z^4 + 15z^2 - 20 + \frac{15}{z^2} - \frac{6}{z^4} + \frac{1}{z^6}$$

$$= \left(z^6 + \frac{1}{z^6}\right) - 6\left(z^4 + \frac{1}{z^4}\right) + 15\left(z^2 + \frac{1}{z^2}\right) - 20$$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$$\left(z - \frac{1}{z}\right)^6 = 2\cos 6\theta - 6 \times 2\cos 4\theta + 15 \times 2\cos 2\theta - 20$$

So $(2\mathrm{i}\sin\theta)^6 = 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$

$-64\sin^6\theta = 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$

$\therefore \sin^6\theta = -\frac{1}{32}\cos 6\theta + \frac{3}{16}\cos 4\theta - \frac{15}{32}\cos 2\theta + \frac{5}{16}$

30) b)

$$\int_0^{\frac{\pi}{4}} \sin^6\theta \,\mathrm{d}\theta = \int_0^{\frac{\pi}{4}} -\frac{1}{32}\cos 6\theta + \frac{3}{16}\cos 4\theta - \frac{15}{32}\cos 2\theta + \frac{5}{16} \,\mathrm{d}\theta$$

$$= \left[-\frac{1}{192}\sin 6\theta + \frac{3}{64}\sin 4\theta - \frac{15}{64}\sin 2\theta + \frac{5}{16}\theta\right]_0^{\frac{\pi}{4}}$$

$$= \left(\frac{1}{192} + 0 - \frac{15}{64} + \frac{5}{16}\times\frac{\pi}{4}\right) - (0)$$

$$= -\frac{11}{48} + \frac{5\pi}{64}$$

$$= \frac{-44 + 15\pi}{192}$$

$\therefore p = 15$, $q = 44$ and $r = 192$

30) c)

$I_6 = \frac{6-1}{6}I_4 - \frac{1}{6\times 2^3} = \frac{5}{6}I_4 - \frac{1}{48}$

$I_4 = \frac{4-1}{4}I_2 - \frac{1}{4\times 2^2} = \frac{3}{4}I_2 - \frac{1}{16}$

$I_2 = \frac{2-1}{2}I_0 - \frac{1}{2\times 2^1} = \frac{1}{2}I_0 - \frac{1}{4}$

Since $I_0 = \int_0^{\frac{\pi}{4}} \sin^0\theta \,\mathrm{d}\theta = \int_0^{\frac{\pi}{4}} 1 \,\mathrm{d}\theta = [\theta]_0^{\frac{\pi}{4}} = \frac{\pi}{4}$ then

$I_2 = \frac{\pi}{8} - \frac{1}{4}$

$\Rightarrow I_4 = \frac{3}{4}\left(\frac{\pi}{8} - \frac{1}{4}\right) - \frac{1}{16} = \frac{3\pi}{24} - \frac{1}{4}$

$\Rightarrow I_6 = \frac{5}{6}\left(\frac{3\pi}{24} - \frac{1}{4}\right) - \frac{1}{48} = \frac{15\pi}{192} - \frac{11}{48} = \frac{15\pi - 44}{192}$ which is the same result as before.

31) a)

$z^n = \cos n\theta + \mathrm{i}\sin n\theta$

$z^{-n} = \cos(-n\theta) + \mathrm{i}\sin(-n\theta) = \cos n\theta - \mathrm{i}\sin n\theta$

$z^n + z^{-n} = \cos n\theta + \mathrm{i}\sin n\theta + \cos n\theta - \mathrm{i}\sin n\theta = 2\cos n\theta$

$\therefore z^n + \frac{1}{z^n} = 2\cos n\theta$ as req.

31) b)

Let $\mathrm{f}(x) = 8x^3 - 12x^2 + 6x - 1$

$\mathrm{f}\left(\frac{1}{2}\right) = 1 - 3 + 3 - 1 = 0 \therefore (2x-1)$ is a factor of $8x^3 - 12x^2 + 6x - 1$

31) c)

$\left(z+\frac{1}{z}\right)^3 = z^3+3z+\frac{3}{z}+\frac{1}{z^3} = \left(z^3+\frac{1}{z^3}\right)+3\left(z+\frac{1}{z}\right)$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$\left(z+\frac{1}{z}\right)^3 = 2\cos 3\theta + 3\times 2\cos\theta$

So $(2\cos\theta)^3 = 2\cos 3\theta + 6\cos\theta$

$\therefore 8\cos^3\theta = 2\cos 3\theta + 6\cos\theta$ as req.

31) d)

$z^6-3z^5+6z^4-7z^3+6z^2-3z+1=0$

$(z^6+1)-3(z^5+z)+6(z^4+z^2)-7z^3=0$

$\left(z^3+\frac{1}{z^3}\right)-3\left(z^2+\frac{1}{z^2}\right)+6\left(z+\frac{1}{z}\right)-7=0$

$(2\cos 3\theta)-3(2\cos 2\theta)+6(2\cos\theta)-7=0$

$2\cos 3\theta+6\cos\theta-6\cos 2\theta+6\cos\theta-7=0$

$8\cos^3\theta-6(2\cos^2\theta-1)+6\cos\theta-7=0$

$8\cos^3\theta-12\cos^2\theta+6+6\cos\theta-7=0$

$8\cos^3\theta-12\cos^2\theta+6\cos\theta-1=0$

$(2\cos\theta-1)^3=0$

$\Rightarrow \cos\theta=\frac{1}{2}$

$\theta=\frac{\pi}{3}$ or $\theta=\frac{5\pi}{3}$

gives $z=\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}=\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}$ and $z=\cos\frac{5\pi}{3}+\mathrm{i}\sin\frac{5\pi}{3}=\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}$

so the solutions are $\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}$ and $\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}$

32) a)

$z^n=\cos n\theta+\mathrm{i}\sin n\theta$

$z^{-n}=\cos(-n\theta)+\mathrm{i}\sin(-n\theta)=\cos n\theta-\mathrm{i}\sin n\theta$

$z^n+z^{-n}=\cos n\theta+\mathrm{i}\sin n\theta+\cos n\theta-\mathrm{i}\sin n\theta=2\cos n\theta$

$\therefore z^n+\frac{1}{z^n}=2\cos n\theta$ as req.

32) b)

$\left(z+\frac{1}{z}\right)^3 = z^3+3z+\frac{3}{z}+\frac{1}{z^3} = \left(z^3+\frac{1}{z^3}\right)+3\left(z+\frac{1}{z}\right)$

Since $z^n+z^{-n} = 2\cos n\theta$ then

$\left(z+\frac{1}{z}\right)^3 = 2\cos 3\theta + 3\times 2\cos\theta$

So $(2\cos\theta)^3 = 2\cos 3\theta + 6\cos\theta$

$\therefore \cos^3\theta = \frac{1}{4}\cos 3\theta + \frac{3}{4}\cos\theta = \frac{1}{4}(\cos 3\theta + 3\cos\theta)$ as req.

32) c)

$z^6+3z^4+z^3+3z^2+1=0$

$(z^6+1)+3(z^4+z^2)+z^3=0$

$\left(z^3+\frac{1}{z^3}\right)+3\left(z+\frac{1}{z}\right)+1=0$

$2\cos(3\theta)+6\cos\theta+2=0$

$8\cos^3\theta+1=0$

$\cos^3\theta = -\frac{1}{8}$

$\cos\theta = -\frac{1}{2}$

$\theta = \frac{2\pi}{3}$ or $\theta = \frac{4\pi}{3}$

gives $z = \cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3} = -\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}$ and $z = \cos\frac{4\pi}{3}+\mathrm{i}\sin\frac{4\pi}{3} = -\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}$

$\left(z-\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}\right)\right)\left(z-\left(-\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}\right)\right) = z^2+z+1$

$z^6+3z^4+z^3+3z^2+1 = (z^2+z+1)(z^4-z^3+3z^2-z+1)$

33) a)

$\left(z+\frac{1}{z}\right)^5 = z^5+5z^3+10z+\frac{10}{z}+\frac{5}{z^3}+\frac{1}{z^5} = \left(z^5+\frac{1}{z^5}\right)+5\left(z^3+\frac{1}{z^3}\right)+10\left(z+\frac{1}{z}\right)$

Since $z^n+z^{-n} = 2\cos n\theta$ then

$\left(z+\frac{1}{z}\right)^5 = 2\cos 5\theta + 5\times 2\cos 3\theta + 10\times 2\cos\theta$

So $(2\cos\theta)^5 = 2\cos 5\theta + 10\cos 3\theta + 20\cos\theta$

$\therefore 32\cos^5\theta = 2\cos 5\theta + 10\cos 3\theta + 20\cos\theta$

33) b)

$z^{10} + 5z^8 + 10z^6 - z^5 + 10z^4 + 5z^2 + 1 = 0$

$(z^{10} + 1) + 5(z^8 + z^2) + 10(z^6 + z^4) - z^5 = 0$

$\left(z^5 + \frac{1}{z^5}\right) + 5\left(z^3 + \frac{1}{z^3}\right) + 10\left(z + \frac{1}{z}\right) - 1 = 0$

$2\cos 5\theta + 5(2\cos 3\theta) + 10(2\cos\theta) - 1 = 0$

$2\cos 5\theta + 10\cos 3\theta + 20\cos\theta - 1 = 0$

$32\cos^5\theta = 1$

$\cos^5\theta = \frac{1}{32}$

$\cos\theta = \frac{1}{2}$

$\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

gives $z = \cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3} = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$ or $z = \cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3} = \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

$\left(z - \left(\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}\right)\right)\left(z - \left(\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right)\right) = z^2 - z + 1$

$z^{10} + 5z^8 + 10z^6 - z^5 + 10z^4 + 5z^2 + 1$

$= (z^2 - z + 1)(z^8 + z^7 + 5z^6 + 4z^5 + 9z^4 + 4z^3 + 5z^2 + z + 1)$

34)

$25z^4 - 10z^3 + 2z^2 - 10z + 25 = 0$

$25(z^4 + 1) - 10(z^3 + z) + 2z^2 = 0$

$25\left(z^2 + \frac{1}{z^2}\right) - 10\left(z + \frac{1}{z}\right) + 2 = 0$

$25(2\cos(2\theta)) - 10(2\cos\theta) + 2 = 0$

$25\cos(2\theta) - 10\cos\theta + 1 = 0$

$25(2\cos^2\theta - 1) - 10\cos\theta + 1 = 0$

$50\cos^2\theta - 25 - 10\cos\theta + 1 = 0$

$25\cos^2\theta - 5\cos\theta - 12 = 0$

$(5\cos\theta + 3)(5\cos\theta - 4) = 0$

Solving gives:

$\cos\theta = -\frac{3}{5} \Rightarrow \sin\theta = \pm\frac{4}{5}$

$\cos\theta = \frac{4}{5} \Rightarrow \sin\theta = \pm\frac{3}{5}$

$\therefore z = -\frac{3}{5} + \frac{4}{5}\mathrm{i}$ or $z = -\frac{3}{5} - \frac{4}{5}\mathrm{i}$ or $z = \frac{4}{5} - \frac{3}{5}\mathrm{i}$ or $z = \frac{4}{5} + \frac{3}{5}\mathrm{i}$